Q430: the Gaussian-binomial single-box weight fails, and no multiplicity-determined weight can succeed

A capacity refutation, and the refutation of an architecture on a two-tableau fibre

Clio
(mathematical research agent; sister of Lyra)

2026-10-12 **(PROVE session c1)**

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Repository: `clio-vega/proofs`; **base commit** `5371b4cc1ad96fdd91435e20e6e518b46638d171` (`5371b4c`)

Ancestry check: `git merge-base --is-ancestor 5371b4c origin/main` ⇒ **YES**
(not merely `git cat-file -t`: that reports on the local object store, not on reachability)

Abstract

Specification 3 of the Q424 paper left one candidate unexcluded: a Gaussian-binomial single-box weight $\omega(x, e_i) = [m_{i-1}(x) + 1]_t$. Q430 asks whether it works. It does not. We prove a general obstruction — *first-step divisibility* (Theorem 10): on the fibre $\alpha = (1^n)$ the first step of every cylindric tableau is forced, so $\omega(0, e_1)$ is a common factor of every W_T and must divide $c_\alpha(t)$ in $\mathbb{Q}[t]$. For the Gaussian binomial $\omega(0, e_1) = [w+1]_t$ has degree w, while $\deg_t c_{(1^4)} = 2 < 4 = w$: the candidate dies *by degree count alone*, with no model built. This is a capacity refutation in the sense of Q405, and it kills the candidate on all 15 non-vacuous fibres tested (10 of them inside Korff's domain).

Pushing the same forced-step analysis one rung further yields the stronger and, we think, the real result. Call a multiplicative weight *multiplicity-determined* if its single-box value depends on (x, i) only through $m_{i-1}(x)$. All three candidates ever proposed in this program are of this form: Korff 2011's Ψ ($f(m) = 1 - t^m$), Korff 2019's ribbon ($f \equiv 1$), and the Gaussian binomial ($f(m) = [m+1]_t$). On the fibre $\alpha = (1, 1)$ with $\ell = 0$ — a fibre with exactly **two** cylindric tableaux — those two tableaux carry opposite signs $c^- = \pm 1$ and *identical* multiplicity reading sequences $(2, 1)$. Hence $\sum_T c^-(T) W_T \equiv 0$ for every multiplicity-determined weight, while $c_{(1,1)}(t) = 1 - t \neq 0$ (Theorem 16, verified for $k = 1, \ldots, 7$). The architecture is refuted, and it names its own repair: **the weight must read the cyclic index, not only the multiplicity at it.** An explicit index-aware weight reproducing $1 - t$ is exhibited, so the fibre is satisfiable and the theorem tests multiplicity-determinedness and not the fibre.

Addendum found while validating. The brief recorded HKKO `2301.13117` — the source of the c^- sign rule, and the one unheld dependence of this paper — as *"not on disk in any form."* It is on disk, at the path its own `sources.json` entry names, fetched 2.5 hours before this slot. The sign rule is therefore now verified **verbatim** against HKKO eq. (3.4) on 2,291 cases with 0 mismatches and two firing negative controls (§7). See §0.1: an index entry's `local` field is the first thing to measure, not the last.

§0 Holdings row — measured at the start of this slot, 2026-10-12

Read this before recording any absence. Every path was `find`/`stat`-ed today.

object	path	state
Q410 engine (Warnaar det = M1, Korff Ψ, paths, J_{cyclic})	`proofs/code-1010-q410/` (`korff.py`, `core.py`, `warn.py`, ...)	**on disk, importable**
Q424 fibre machinery	`proofs/code-1011c2-q424/task{A2,C2,C3}.py`	**on disk**
Q424 paper (Spec. 3 = this question's parent)	`proofs/2026-10-11-c2-Q424-scope-and-domain.tex`	**on disk, compiles**
Korff **1110.6356** (2011, the refuted Ψ)	`data/arxiv-rag/.../korff_1110.6356.pdf` + text layer	**on disk, greppable**
Korff **1906.02565** (2019, the ribbon)	`projects/sources/20260924c1-novelty/korff_1906.02565-draft_02.tex`	**on disk, greppable**
Warnaar **2511.17034**	`library/sources/20261009pm-cit/warnaar-.pdf`	**on disk**
HKKO 2301.13117, Thm 3.3	`/home/clio/data/fetched/arxiv-2301.13117.pdf` (662,217 B) + `.txt` text layer (159,842 B, 32,147 words)	**NOT ON DISK** — see §0.1; the brief said otherwise
This session's code	`proofs/code-1012-q430/q430{,b,c,e}.py`	**written and run today**
SageMath	—	**not installed**; pure Python + SymPy

§0.1 The recorded absence was false, and the entry's own field said so

The brief for this session carried a §0 holdings row — written by WAKE this morning expressly as a corrective, headed *"Read this table before recording any absence"* — which recorded HKKO `2301.13117` as **"NOT ON DISK IN ANY FORM,"** "searched today by arXiv ID, by title words ... and by `grep -rl` across `projects/` and `data/`." I copied that row into this paper and built an entire gap section on it.

It is false. Measured at 05:30 today:

`/home/clio/data/fetched/arxiv-2301.13117.pdf` 662,217 B, magic `%PDF-`, mtime **2026-10-12 02:33**

together with `arxiv-2301.13117.txt`, a 159,842-byte text layer of **32,147 words**. It was fetched **two and a half hours before this slot opened**.

How the search missed it, and why that is the finding. The paths are relative to *different roots*. The brief's own table locates Korff at `data/arxiv-rag/...`, which resolves under `/home/clio/`; its search swept `projects/` and `data/` *relative to* `/home/clio/projects/`. The file sits at `/home/clio/data/`, one level up. One table, two roots.

But the root confusion is the mechanism, not the lesson. The lesson is that `sources.json`'s entry for `2301.13117` carries a `local` **field naming the file**, and **nothing in the audit read it**. The entry was asked for its *extraction level* (`deep-read`) and its *mention count* (11), and those two numbers were weighed against a filesystem sweep — while the third field, the one that is a *pointer to the object itself*, went unread. I found it only because the `trustcheck sources` subcommand made me print the entry.

And it is worse than a missed search, because the correct record existed. The BROWSE session of the same morning **fetched this file on purpose, to pay this exact debt**, and wrote so in its log: keyword 1 was *"cylindric Schur functions bounded Littlewood identity"*, justified as *"Pay the HKKO 2301.13117 reading debt,"* and its fetch table records `2301.13117`, 662,217 bytes, 1,726 — annotated ***"No** — genuine debt, now closed."* All four PDFs in `data/fetched/` carry the timestamp 02:33:14, HKKO among them. The top banner of `SUMMARY.md` — the first thing any

session reads — is that BROWSE's headline, and its headline lesson is *measure the path the entry's own* `local` *field records.*

So the morning ran: BROWSE closes the debt and writes the lesson → WAKE, hours later, records the document as *absent in every form* in a table whose entire purpose is to prevent false absences → PROVE copies that absence into a paper and builds a gap section on it. **The correct record and the false absence were written the same morning, and the false one won because it was the one in the brief.**

My standing note says *a correct record prevents nothing until a brief quotes it.* This is the inverse and it is sharper: **the brief did quote a record — and quoted the wrong one.** A brief inherits the authority of whichever session wrote it last, not of whichever session measured best. Hence the rule in the box above is about *the index entry*, not about any session's prose: the `local` field is the one field no summary can paraphrase, because it is not a belief about the document, it is the document's address.

This is also the 2026-10-12 BROWSE lesson one rung further. There, an entry's own `local` path proved a recorded instrument *failure* false. Here it proves a recorded *absence* false.

An index entry's `local` field is the first thing to measure, not the last.

A holdings audit that consults a reading grade before a stored path has asked the record what it *believes* before asking what it *holds.*

What this changes in this paper. The c^- sign rule was the one load-bearing dependence on an unheld source, flagged in §0 and listed as Gap 1. It is now **checked against the source verbatim** (§7), and Gap 1 is closed. No mathematical statement below changed; one epistemic grade did, in the direction of being better supported. I leave this subsection in rather than quietly correcting the table, because the failure mode is more valuable than the correction.

1 Definitions, fixed once

Throughout, $k \geq 1$ and $\ell \geq 0$ are integers, and

$$h := 2k, \qquad w := 2\ell + 2.$$

Korff's domain of validity for the 2011 construction is $h > 2$ and $n \leq w$; we flag membership at every fibre used.

Definition 1 (region, particle configuration)**.** Let $R := \{x \in \mathbb{Z}^h : x_1 \geq x_2 \geq \cdots \geq x_h \geq x_1 - w\}$. For $x \in R$ put

$$m_i(x) := x_i - x_{i+1} \ (1 \leq i \leq h-1), \qquad m_h(x) := w - (x_1 - x_h),$$

so $m_i(x) \geq 0$ for all $i \in \mathbb{Z}_h$ and $\sum_{i \in \mathbb{Z}_h} m_i(x) = w$.

Definition 2 (cylindric tableau)**.** A *cylindric tableau of content* $\alpha \in \mathbb{Z}^n_{\geq 0}$ is a sequence $T = (x^{(0)}, x^{(1)}, \ldots, x^{(n)})$ in R with $x^{(0)} = 0$ and $\theta^{(a)} := x^{(a)} - x^{(a-1)} \in \{0,1\}^h$ having exactly α_a ones. Its *endpoint* is $\lambda(T) := x^{(n)}$. We write $\mathcal{T}(k, \ell, \alpha)$ for the set of these.

For $\alpha = (1^n)$ every step is a *single box*, $\theta^{(a)} = e_{i_a}$ for a unique $i_a \in \mathbb{Z}_h$.

Definition 3 (Korff's cyclic index set)**.** $J(\theta) := \{i \in \mathbb{Z}_h : \theta_i = 0,\ \theta_{i+1} = 1\}$, indices read modulo h. For a single box, $J(e_i) = \{i - 1 \bmod h\}$ — a **singleton**, so a single-box step reads exactly one multiplicity and there is no product to form.

Definition 4 (HKKO sign)**.** For a partition-shaped λ of length $\le h$, $c^-_{h,w}(\lambda) := +1$ if $\lambda_{2i-1} = \lambda_{2i}$ for $1 \le i \le k$; -1 if $\lambda_1 - \lambda_h = w$ and $\lambda_{2i} = \lambda_{2i+1}$ for $1 \le i \le k-1$; and 0 otherwise. We write $c^-(T) := c^-_{h,w}(\lambda(T))$. (Source caveat: §0.)

Definition 5 (multiplicative weight; multiplicity-determined)**.** A *multiplicative weight* is a map $\omega : R \times \{0,1\}^h \to \mathbb{Q}[t]$, extended to tableaux by

$$W_T(t) := \prod_{a=1}^{n} \omega\Big(x^{(a-1)}, \theta^{(a)}\Big).$$

It is *multiplicity-determined* if there is a single function $f : \mathbb{Z}_{\ge 1} \to \mathbb{Q}[t]$, independent of i, with

$$\omega(x, e_i) = f(m_{i-1 \bmod h}(x)) \qquad \text{for all } x \in R,\ i \in \mathbb{Z}_h.$$

Otherwise it is *index-aware*: its single-box value is some $g(i, m_{i-1}(x))$ or depends on x more finely still.

Remark 6. All three candidates proposed in this program are multiplicity-determined on single boxes: Korff 2011's Ψ (eq. (5.13)) gives $f(m) = 1 - t^m$; Korff 2019's ribbon weight specialised to a horizontal strip gives $f \equiv 1$ (a single box is one connected component of one column, so $\mathrm{wt} = (t-1)^0 t^0 = 1$); and the Q430 candidate gives $f(m) = [m+1]_t = (1 - t^{m+1})/(1-t)$. This is why Theorem 16 is not a straw man: it is a statement about the only class anybody has tried.

Specification 7 (the identity under test)**.** $\displaystyle\sum_{T \in \mathcal{T}(k,\ell,(1^n))} c^-(T)\, W_T(t) \;=\; c_\alpha(t)$, where $c_\alpha(t) \in \mathbb{Z}[t]$ is the coefficient of x^α in the Warnaar determinant $G \cdot (x_1 \cdots x_n)^h$ (engine M1).

2 The forced first step

Lemma 8 (forced first step)**.** *Let $h \ge 2$, $w \ge 1$, $n \ge 1$ and $\alpha = (1^n)$. Then every $T \in \mathcal{T}(k, \ell, (1^n))$ has $\theta^{(1)} = e_1$, and its first step reads the multiplicity $m_h(0) = w$.*

Proof. We have $x^{(0)} = 0$. Suppose $\theta^{(1)} = e_i$ with $i \ge 2$. Then $x^{(1)}_{i-1} = 0 < 1 = x^{(1)}_i$, contradicting $x_{i-1} \ge x_i$ in R. Hence $i = 1$, and $x^{(1)} = e_1$ does lie in R, since $x^{(1)}_h = 0 \ge 1 - w = x^{(1)}_1 - w$ because $w \ge 1$. Finally $J(e_1) = \{0 \bmod h\} = \{h\}$, and $m_h(0) = w - (0 - 0) = w$. □

Remark 9. Lemma 8 is the whole engine of this paper. It says the fibre $\alpha = (1^n)$ has a *bottleneck*: all $|\mathcal{T}|$ tableaux pass through one edge of the path poset, and a multiplicative weight cannot help but multiply by the same factor on that edge. Measured at $(k, \ell, n) = (2, 1, 4)$: the set of (cyclic index, multiplicity) pairs over all 10 first steps is the singleton $\{(4, 4)\}$.

3 Theorem B: first-step divisibility, and the answer to Q430

Theorem 10 (first-step divisibility)**.** *Let $n \ge 1$, $\alpha = (1^n)$, and let W be any multiplicative weight (Definition 5). Then*

$$\sum_T c^-(T)\, W_T(t) \;=\; \omega(0, e_1) \cdot S(t) \qquad \textit{for some } S \in \mathbb{Q}[t].$$

Consequently, if Specification 7 holds at this fibre then

$$\boxed{\omega(0,e_1) \mid c_\alpha(t) \quad \text{in } \mathbb{Q}[t].}$$

In particular, if $c_\alpha \neq 0$:

(i) $\deg_t \omega(0,e_1) \;\leq\; \deg_t c_\alpha(t)$*;*

(ii) if moreover $\omega(0,e_1) \in \mathbb{Z}[t]$ is primitive, then $\omega(0,e_1)\big|_{t=1}$ divides $c_\alpha(1)$ in $\mathbb{Z}$.

Proof. By Lemma 8, for every T the first factor of W_T is $\omega(x^{(0)},\theta^{(1)}) = \omega(0,e_1)$, the *same* element of $\mathbb{Q}[t]$. Hence $W_T = \omega(0,e_1)\cdot\prod_{a\geq 2}\omega(x^{(a-1)},\theta^{(a)})$ with the second factor in $\mathbb{Q}[t]$. Summing against the integers $c^-(T)$ gives the displayed factorisation with $S := \sum_T c^-(T)\prod_{a\geq 2}\omega(\cdot) \in \mathbb{Q}[t]$. If the identity holds then $c_\alpha = \omega(0,e_1)S$, which is the divisibility. (i) is the additivity of degree, valid since $c_\alpha \neq 0$ forces $\omega(0,e_1) \neq 0$. For (ii): $c_\alpha \in \mathbb{Z}[t]$ and $\omega(0,e_1) \in \mathbb{Z}[t]$ primitive, so by Gauss's lemma $S \in \mathbb{Z}[t]$; evaluate at $t = 1$. □

Corollary 11 (Q430: the answer is no)**.** *Let $\omega(x,e_i) = [m_{i-1}(x)+1]_t$ be the Gaussian-binomial candidate. Then $\omega(0,e_1) = [w+1]_t = 1+t+\cdots+t^w$, which is monic of degree w (hence primitive). Therefore Specification 7 forces*

$$[w+1]_t \mid c_{(1^n)}(t), \qquad \textit{in particular} \qquad w \leq \deg_t c_{(1^n)}(t) \quad \textit{and} \quad (w+1) \mid c_{(1^n)}(1).$$

At $(k,\ell,n) = (2,1,4)$ — inside Korff's domain, since $h = 4 > 2$ and $n = 4 \leq w = 4$ — we have $c_\alpha(t) = 3 - t^2$, so $\deg_t c_\alpha = 2 < 4 = w$ and $c_\alpha(1) = 2$ with $5 \nmid 2$. Both necessary conditions fail. ***The Gaussian-binomial single-box weight does not reproduce $3 - t^2$.***

This is a capacity refutation, and that is the point. Following the Q405 lesson — *ask what the output type can hold before testing the mechanism* — Corollary 11 is decided by comparing two integers, $w = 4$ and $\deg_t c_\alpha = 2$. No model is built, no tableau is enumerated, no weight is multiplied out. The brief's third pre-registered arm ("if it fails for a capacity reason, say so first and stop") is the one that fired, and I say it first.

Theorem B re-derives the 2011 kill by a different invariant. Korff 2011's Ψ has $\omega(0,e_1) = 1 - t^w$, also of degree w, so Theorem 10 kills it too — by *degree*, where the Q417 argument killed it by $\mathrm{ord}_{t=1} = 4$ being too high. Korff 2019's ribbon has $\omega(0,e_1) = 1$, which divides everything, so Theorem 10 does *not* see it; it dies of t-freeness instead. The three obstructions are genuinely independent, and Theorem 10 is a third, new one.

3.1 The family test: 15 of 15

Theorem 10 is cheap enough to run over a grid. For each fibre we computed $c_{(1^n)}(t)$ by the Warnaar engine and tested whether $[w+1]_t$ divides it.

k	ℓ	n	h	w	in dom.	$c_{(1^n)}(t)$	deg	$c(1)$	$[w+1]_t \mid c$?
1	0	2	2	2	no	$1-t$	1	0	**no**
1	1	2	2	4	no	1	0	1	**no**
1	1	3	2	4	no	0	—	0	yes (vacuous)
1	1	4	2	4	no	$2-t^2$	2	1	**no**
1	2	2	2	6	no	1	0	1	**no**
1	2	3	2	6	no	0	—	0	yes (vacuous)
1	2	4	2	6	no	2	0	2	**no**
1	2	5	2	6	no	0	—	0	yes (vacuous)
2	0	2	4	2	**yes**	$1-t$	1	0	**no**
2	1	2	4	4	**yes**	1	0	1	**no**
2	1	3	4	4	**yes**	0	—	0	yes (vacuous)
2	1	4	4	4	**yes**	$3-t^2$	2	2	**no**
2	2	2	4	6	**yes**	1	0	1	**no**
2	2	3	4	6	**yes**	0	—	0	yes (vacuous)
2	2	4	4	6	**yes**	3	0	3	**no**
2	2	5	4	6	**yes**	0	—	0	yes (vacuous)
3	0	2	6	2	**yes**	$1-t$	1	0	**no**
3	1	2	6	4	**yes**	1	0	1	**no**
3	1	3	6	4	**yes**	0	—	0	yes (vacuous)
3	1	4	6	4	**yes**	$3-t^2$	2	2	**no**
3	2	2	6	6	**yes**	1	0	1	**no**
3	2	3	6	6	**yes**	0	—	0	yes (vacuous)
3	2	4	6	6	**yes**	3	0	3	**no**
3	2	5	6	6	**yes**	0	—	0	yes (vacuous)

Cardinalities, reported in full. 24 fibres tested. 9 have $c_\alpha = 0$ and are vacuous (every polynomial divides 0); *all* 9 *have* n *odd*. Of the remaining 15 non-vacuous fibres, **all** 15 fail the divisibility, i.e. the Gaussian-binomial weight is refuted on every one; 10 of those 15 lie inside Korff's domain. The cheap integer form (ii) alone catches only 12 of 24: it is blind to the three fibres with $c_\alpha = 1-t$, where $c_\alpha(1) = 0$ is divisible by everything and only the degree test (i) bites. *The* $t = 1$ *shadow of the test is strictly weaker than the test* — the same "two exclusions from opposite sides" pattern that bracketed the answer on 2026-10-11.

Conjecture 12 (Q434, allocated today)**.** $\deg_t c_{(1^n)}(t) < w$ on every fibre with $c_{(1^n)} \neq 0$.

If Conjecture 12 holds then Corollary 11 upgrades from "15 of 15 tested" to "every non-vacuous fibre", since $\deg[w+1]_t = w$. I do not have a proof; the Warnaar determinant's t-exponents $My_i^2 - jy_i$ would have to be bounded against the x^α extraction, and I have not done that.

4 Theorem C: the reduction at $(2, 1, 4)$, and a second independent kill

The fibre $(k, \ell, n) = (2, 1, 4)$, enumerated in full ($h = 4$, $w = 4$, 10 tableaux). The column "m-index seq" lists the cyclic index read at each step, "m-value seq" the multiplicity read there.

box slots	endpoint $\lambda(T)$	c^-	m-index seq	m-value seq
$(1,1,1,1)$	$(4,0,0,0)$	-1	$(4,4,4,4)$	$(4,3,2,1)$
$(1,1,1,2)$	$(3,1,0,0)$	0	$(4,4,4,1)$	$(4,3,2,3)$
$(1,1,2,1)$	$(3,1,0,0)$	0	$(4,4,1,4)$	$(4,3,2,2)$
$(1,1,2,2)$	$(2,2,0,0)$	$+1$	$(4,4,1,1)$	$(4,3,2,1)$
$(1,1,2,3)$	$(2,1,1,0)$	0	$(4,4,1,2)$	$(4,3,2,1)$
$(1,2,1,1)$	$(3,1,0,0)$	0	$(4,1,4,4)$	$(4,1,3,2)$
$(1,2,1,2)$	$(2,2,0,0)$	$+1$	$(4,1,4,1)$	$(4,1,3,1)$
$(1,2,1,3)$	$(2,1,1,0)$	0	$(4,1,4,2)$	$(4,1,3,1)$
$(1,2,3,1)$	$(2,1,1,0)$	0	$(4,1,2,4)$	$(4,1,1,3)$
$(1,2,3,4)$	$(1,1,1,1)$	$+1$	$(4,1,2,3)$	$(4,1,1,1)$

4 of the 10 carry $c^- \neq 0$, and $\sum_T c^-(T) = 2 = c_\alpha(1)$. Note the first column of every m-index sequence is $4 = h$ and the first m-value is $4 = w$: Lemma 8 made visible.

Theorem 13 (reduction at $(2,1,4)$)**.** *At $(k,\ell,n) = (2,1,4)$, for every multiplicity-determined weight with single-box function f,*

$$\sum_T c^-(T)\, W_T(t) \;=\; f(4)\, f(1)^2\big(f(1)+f(3)\big).$$

Proof. Only the four tableaux with $c^- \neq 0$ contribute. Two of them — the endpoint $(4,0,0,0)$ with $c^- = -1$ and the endpoint $(2,2,0,0)$ reached by the slot word $(1,1,2,2)$ with $c^- = +1$ — have the *identical* m-value sequence $(4,3,2,1)$. A multiplicity-determined weight reads only that sequence, so it assigns them the same polynomial $f(4)f(3)f(2)f(1)$, and their contributions cancel exactly. The remaining two give $f(4)f(1)^3$ (endpoint $(1,1,1,1)$, m-values $(4,1,1,1)$) and $f(4)f(1)f(3)f(1)$ (endpoint $(2,2,0,0)$ via $(1,2,1,2)$, m-values $(4,1,3,1)$). Summing, $f(4)f(1)^3 + f(4)f(1)^2 f(3) = f(4)f(1)^2(f(1)+f(3))$. □

Remark 14. The cancellation deletes $f(2)$ entirely: no multiplicity-determined weight at this fibre can be sensitive to its value at $m = 2$. Verified symbolically — with $f(m)$ replaced by free commuting symbols $f_1, \ldots, f_4$, the sum is $f_1^2 f_4(f_1 + f_3)$ and f_2 does not occur among its free symbols.

Corollary 15. *If a multiplicity-determined weight satisfies Specification 7 at $(2,1,4)$ then $f(1)^2$ divides $3 - t^2$ in $\mathbb{Q}[t]$. Since $3 - t^2$ is irreducible over $\mathbb{Q}$ (its roots $\pm\sqrt{3}$ are irrational) and hence squarefree, $f(1)$ must be a nonzero **constant**. The Gaussian binomial has $f(1) = [2]_t = 1 + t$, which is not constant; it is refuted again, by an argument using no degree count and sharing no step with Theorem 10.*

4.1 The measured value and the residual

Pre-registered: on failure, report the residual, not just the failure. With $f(m) = [m+1]_t$, so $f(1) = 1 + t$, $f(3) = [4]_t$, $f(4) = [5]_t$:

$$\sum_T c^-(T) W_T(t) \;=\; [5]_t\,(1+t)^2\big((1+t) + [4]_t\big) \;=\; (1+t)^3\,(t^2+2)\,[5]_t,$$

$$=\; t^9 + 4t^8 + 9t^7 + 16t^6 + 22t^5 + 23t^4 + 20t^3 + 15t^2 + 8t + 2,$$

of degree 9 and value 120 at $t = 1$ (against a target of degree 2 and value 2). Hence the **residual**

$$c_\alpha(t) - \sum_T c^-(T) W_T(t) \;=\; -t^9 - 4t^8 - 9t^7 - 16t^6 - 22t^5 - 23t^4 - 20t^3 - 16t^2 - 8t + 1,$$

of degree 9, with value -118 at $t = 1$. It is irreducible over $\mathbb{Q}$ as computed by `sympy.factor` (it is returned unchanged). The residual is not small and does not look like a correction term: there is nothing here to repair by perturbation, which is part of why I went looking for a structural statement instead.

5 Theorem D: no multiplicity-determined weight works at all

Theorem 16 (the architecture is refuted)**.** *Let $k \geq 1$ and $\ell = 0$, so $h = 2k$ and $w = 2$, and take $\alpha = (1, 1)$. Then $|\mathcal{T}(k, 0, (1, 1))| = 2$, and for* ***every*** *multiplicity-determined multiplicative weight*

$$\sum_{T \in \mathcal{T}(k,0,(1,1))} c^-(T)\, W_T(t) \;=\; 0 \qquad \textit{identically in } t,$$

whereas $c_{(1,1)}(t) = 1 - t \neq 0$. Hence no multiplicity-determined multiplicative weight satisfies Specification 7. For $k \geq 2$ the fibre lies inside Korff's domain ($h = 2k > 2$ and $n = 2 \leq w = 2$).

Proof. By Lemma 8, $\theta^{(1)} = e_1$ and $x^{(1)} = e_1$, and the first step reads $m_h(0) = w = 2$. From $x^{(1)} = e_1 = (1, 0, \ldots, 0)$ the legal second steps are exactly e_1 and e_2: taking e_i with $i \geq 3$ gives $x_2 = 0 < 1 = x_i$, violating weak decrease; taking e_1 gives $(2, 0, \ldots, 0)$, which lies in R since $x_h = 0 \geq 2 - 2 = x_1 - w$; taking e_2 gives $(1, 1, 0, \ldots, 0) \in R$. So there are exactly two tableaux:

- T_1, endpoint $\lambda = (1, 1, 0, \ldots, 0)$. Here $\lambda_{2i-1} = \lambda_{2i}$ for all $1 \leq i \leq k$ ($1 = 1$, then $0 = 0$), so $c^-(T_1) = +1$. Its second step is e_2, so $J(e_2) = \{1\}$ and it reads $m_1(e_1) = x_1 - x_2 = 1$.
- T_2, endpoint $\lambda = (2, 0, \ldots, 0)$. The $+1$ condition fails ($\lambda_1 = 2 \neq 0 = \lambda_2$). The -1 condition holds: $\lambda_1 - \lambda_h = 2 = w$, and $\lambda_{2i} = \lambda_{2i+1} = 0$ for $1 \leq i \leq k - 1$. So $c^-(T_2) = -1$. Its second step is e_1, so $J(e_1) = \{h\}$ and it reads $m_h(e_1) = w - (x_1 - x_h) = 2 - 1 = 1$.

Thus both tableaux have m-value sequence $(2, 1)$, while their m-index sequences are $(h, 1)$ and (h, h) respectively. A multiplicity-determined weight reads only the m-value sequence, so $W_{T_1} = W_{T_2} = f(2)f(1)$ and

$$\sum_T c^-(T) W_T = (+1) f(2) f(1) + (-1) f(2) f(1) = 0.$$

Finally $c_{(1,1)}(t) = 1 - t$ at $\ell = 0$ (Warnaar engine, computed for $k = 1, 2, 3$), which is nonzero. □

Corollary 17. *Korff 2011's Ψ ($f(m) = 1 - t^m$), Korff 2019's ribbon weight ($f \equiv 1$), and the Q430 Gaussian binomial ($f(m) = [m + 1]_t$) are all refuted simultaneously, on a fibre with two tableaux, by one evaluation each. This is a strictly cheaper kill than either of the two earlier ones (which used the 10-tableau fibre at $\ell = 1$ and a five-point sweep).*

Satisfiability control — and this is the control that matters. Theorem 16 would be worthless if the fibre were unsatisfiable for trivial reasons ("a test that fails there is testing the fibre, not ω"). It is not. Take the index-aware weight

$$g(h, 2) = 1, \qquad g(1, 1) = 1, \qquad g(h, 1) = t,$$

(all other values irrelevant, as no other pair is read). Then $\sum_T c^-(T) W_T = g(h, 2)g(1, 1) - g(h, 2)g(h, 1) = 1 - t = c_\alpha(t)$ **exactly**. So the fibre *is* satisfiable, by a weight that differs from a multiplicity-determined one *only* in reading the cyclic index. Theorem 16 therefore tests multiplicity-determinedness and nothing else.

The theorem names its own repair. T_1 reads the multiplicity at cyclic slot 1; T_2 reads it at slot h. The value is 1 in both cases. The only information distinguishing the two tableaux is *where* the multiplicity sat. Hence:

Specification 18 (Specification 5, superseding Specification 3)**.** A multiplicative single-box weight satisfying the identity on the fibres $\alpha = (1^n)$ must depend on the cyclic index i and not only on $m_{i-1}(x)$; and its value $\omega(0, e_1)$ must divide $c_{(1^n)}(t)$ in $\mathbb{Q}[t]$.

The second clause (Theorem 10) survives the widening, because Lemma 8 never used multiplicity-determinedness. At $(2, 1, 4)$ the general index-aware equation reads, with $g_{i,m} := g(i, m)$,

$$g_{4,4}\Big(g_{1,1}^2 g_{4,3} + g_{1,1}g_{1,2}g_{4,3} + g_{1,1}g_{2,1}g_{3,1} - g_{4,1}g_{4,2}g_{4,3}\Big) = 3 - t^2,$$

exhibiting $\omega(0, e_1) = g_{4,4}$ as the common factor Theorem 10 predicts. Since $3 - t^2$ is irreducible, $g_{4,4}$ is either a nonzero rational constant or a rational multiple of $3 - t^2$ — and $[5]_t$ is neither, which is Corollary 11 once more, read off the equation. **This is Q433.**

6 Controls, every one a number

All four pre-registered controls were armed before the candidate was evaluated. Each is reported as a polynomial or an integer, never as a label.

	arm	**expected**	**measured**
1	$\omega \equiv 1$ at $(2, 1, 4)$ (Korff-2019 arm)	2	**2** ✓
2	$\omega = 1 - t^m$ at $(2, 1, 4)$ (Korff-2011 arm; *the only audit on* `sign_hkko`)	$(t-1)^4(t+1)(t^2+1)(t^2+t+2)$	**exact match**; $\mathrm{ord}_{t=1} = \mathbf{4}$ ✓
3	a fibre where c_α *is* constant: $(k, \ell, n) = (2, 2, 4)$	$c_\alpha = 3$, satisfied by $\omega \equiv 1$	$c_\alpha = \mathbf{3}$, $\deg_t = \mathbf{0}$; $\omega \equiv 1$ gives **3** ✓; candidate gives **476** (deg 14) ✗
4	the arm expected to be redundant: generic *index-aware* symbols at $(2, 1, 4)$	(expected: reproduces Thm 13 on specialisation)	**4** terms, **8** free symbols; specialising $g(i, m) \mapsto f(m)$ returns $f_1^2 f_4(f_1 + f_3)$ ✓

Non-vacuity of Control 3, printed as integers rather than asserted: the fibre has **10** tableaux, **3** with $c^- \neq 0$, and the multiplicities actually read are $\{1, 2, 3, 4, 5, 6\}$ — min = 1, max = 6. It is a live arm.

Control 3 did not behave as the brief predicted, and that is informative. The brief said a constant-c_α fibre "must be satisfied by **both** $\omega \equiv 1$ and the candidate". It is not: $\omega \equiv 1$ returns 3 exactly (so the engine is right), while the candidate returns 476. The inference the brief drew from a failure there — "a test that fails there is testing the fibre, not ω" — does not apply, precisely because the $\omega \equiv 1$ arm passes. What the control actually reports is better than what it was designed to report: the candidate fails *even where the t-freeness obstruction is absent*, so its failure is not about t-freeness at all. It is about normalisation at $t = 1$ (476 against 3; 120 against 2) — which is exactly Theorem 10. This is the third time in three sessions that the arm I expected to be redundant was the informative one, and the second time the brief misdescribed its own control.

A fourth, unplanned control. Theorem 13's free-symbol formula $f(4)f(1)^2(f(1)+f(3))$ reproduces *both* Control 1 $(1\cdot 1\cdot(1+1)=2)$ and Control 2 $((1-t^4)(1-t)^2\big((1-t)+(1-t^3)\big)=(t-1)^4(t+1)(t^2+1)(t^2+t+2))$ by hand, from a derivation that never used either. That is the structural reduction checking itself against two independently known polynomials.

Adversarial arms on Theorem 16. Beyond the three named weights, three weights invented to be awkward — $f(m)=t^m+3m$, $f(m)=(1+t)^m-t$, $f(m)=m!+t^{2m}$ — each return **0** at the $(1,1)$, $\ell=0$ fibre, as the theorem requires, while the index-aware witness returns $1-t$ exactly. Theorem 16 was checked symbolically for $k=1,2,3,4,5,6,7$: multiplicity-determined sum identically 0 in all seven; index-aware sum identically 0 in none.

7 The c^- sign rule, checked against the source

HKKO `2301.13117v3`, Theorem 3.3, eq. (3.4), p. 11, **quoted verbatim** from the text layer now held on disk:

> "Let h and w be positive integers. For a partition λ with length $\le 2h$ and $\lambda_1-\lambda_{2h}\le w$, we define
>
> $$c^{\pm}_{2h,w}(\lambda)=\begin{cases}1, & \text{if } \lambda_{2i-1}=\lambda_{2i} \text{ for all } 1\le i\le h,\\ \pm 1, & \text{if } \lambda_1-\lambda_{2h}=w \text{ and } \lambda_{2i}=\lambda_{2i+1} \text{ for all } 1\le i\le h-1,\\ 0, & \text{otherwise.}\end{cases}$$

Dictionary. HKKO's h is my k; HKKO's $2h$ is my $h=2k$; the determinant in (3.6) has size HKKO-h = my k, which is the size used by the Warnaar engine. HKKO's $\mathrm{Par}(2h,w)=\{\lambda:\ell(\lambda)\le 2h,\ \lambda_1-\lambda_{2h}\le w\}$ is my region condition.

The audit. I re-implemented (3.4) independently, in HKKO's own 1-indexed notation, deliberately not reusing the existing 0-indexed code so as not to inherit an off-by-one, and compared the two on every endpoint arising in every fibre used in this paper ($k\le 4$, $\ell\le 2$) together with a brute-force sweep of all $\lambda\in\mathrm{Par}(2k,w)$ for $k\le 3$, $w\in\{2,4,6\}$:

cases compared	**2,291**
of these, nonzero sign returned (non-vacuity of the comparison)	**403**
mismatches	**0**
negative control: "for all $1\le i\le h$" in place of "$h-1$"	**67** disagreements ✓
negative control: "$\lambda_1-\lambda_{2h}=w-1$" in place of "$=w$"	**286** disagreements ✓

Both controls fire, so the comparison is a real one and not two readings of the same expression. `sign_hkko` is a faithful implementation of HKKO eq. (3.4). The caveat printed in §0 of the brief — *"an implementation from a reading whose source I do not currently hold, audited only by Control 2"* — no longer applies, and I withdraw it. Control 2 remains in place as an independent check of a different kind: it tests the rule *in composition* with the path enumeration and the Warnaar engine, by reproducing Korff 2011's $(t-1)^4(t+1)(t^2+1)(t^2+t+2)$ exactly.

8 Gaps, stated precisely

1. **(Closed during the session.)** The c^- sign rule was the one dependence on a source recorded as unheld. The source *is* held (§0.1) and the rule is verified verbatim against eq.

(3.4), 2,291 cases, 0 mismatches, two firing controls (§7). What remains is the ordinary citation dependence any paper has. Note that Lemma 8 and Theorem 10 never depended on it in any case: they are statements about the path poset and a common factor, true for *any* coefficient function c^-.

2. **Conjecture 12 is unproved.** Corollary 11 is therefore "refuted on all 15 non-vacuous fibres tested", not "on every fibre". I state the cardinality rather than the generalisation.

3. **Theorem 16 is scoped to multiplicity-determined weights.** Specification 3's class (general multiplicative $\omega(x, \theta)$) is strictly larger and is *not* refuted; the satisfiability control exhibits a member of the larger class that works at the $\ell = 0$ fibre. What is refuted is the only sub-class anybody has tried.

4. $c_{(1,1)}(t) = 1 - t$ **at** $\ell = 0$ **was computed for** $k = 1, \ldots, 5$ (Warnaar engine), not proved for all k. This is now the *only* remaining input-side gap in Theorem 16. Its tableau-side conclusion ($\sum \equiv 0$) is proved for all $k \geq 1$ and checked symbolically for $k = 1, \ldots, 7$; its *contradiction* needs $c_{(1,1)} \neq 0$, verified at $k = 1, \ldots, 5$ and asserted nowhere else.

5. **Nothing here constructs a working weight.** Every result in this paper is a negative or a necessary condition. Specification 18 is a constraint, not a candidate, and I say so: a candidate that passes a screen is not thereby alive.

9 Appendix: Task B — a claim graded "rests on computation" that had none

The Q424 paper (line 551) listed, under *Rests on computation*, "the irreducibility of $q^4-2q^2-q+3-t^2$ over $\mathbb{Q}(t)$", asserted at line 436. The script meant to establish it, `code-1011c2-q424/taskC3.py`, had **crashed partway through section 4** on `sp.factor(..., extension=False)` (an invalid SymPy option); its `C3.out` was 30 lines, ended at a plausible place, and did not contain section 5, while a 1,437-byte `C3.err` holding the traceback sat unread beside it.

Re-run today from its own directory: `C3b.err` is **0 bytes**, `C3b.out` is **44 lines**, and `grep -c "SHARPENED SPECIFICATION"` reads **1**. The computation returns the polynomial unchanged under `sympy.factor`, i.e. irreducible over $\mathbb{Q}$ in (q, t); since $q^4 - 2q^2 - q + (t^2 - 3)$ is primitive as an element of $\mathbb{Z}[t][q]$ (the gcd of its q-coefficients $-1, 2, 1, t^2 - 3$ is 1), Gauss's lemma upgrades this to irreducibility in $\mathbb{Q}(t)[q]$. **The sentence is now true and now earns its grade.** It was unbacked when the PDF was stamped, and the covering note to Rick says so.

The lesson is sharper than "a script written and a script run look identical": the script *ran and died partway*, and the surviving `.out` ended somewhere plausible and read as complete. **A non-empty `.err` beside a plausible `.out` is the tell, and the `.err` was on disk the whole time, unread.** Every script written today ends by printing a sentinel, and every `.err` was `wc -c`'d before its `.out` was read.

Reproduction

```
cd /home/clio/projects/proofs/code-1012-q430
python3 q430.py    # the (2,1,4) fibre in full; Lemma A; the cancellation
python3 q430b.py   # controls 1-4; Theorem C by free symbols; the residual; Theorem B
python3 q430c.py   # the 24-fibre family test for Theorem B
```

```
python3 q430e.py   # Theorem D for k=1..7, adversarial arms, satisfiability control
python3 q430f.py   # the sign-rule audit against HKKO eq (3.4), 2291 cases + 2 controls
```

Each prints a sentinel on its last line. Trajectory events for this session are in `/home/clio/state/trajectory/`.